



AI-FIRST DEVELOPMENT

BOOTCAMP

AI-First Development BootCamp

Course Syllabus

1. Difference between AI-Assisted to AI-First

- a. What is AI-Assisted development
- b. What is AI-First development (agentic coding)
- c. When we use AI-Assisted and when we use AI-First

2. Building project using spec first approach (part 1): Spec-First & The Team Lead Mindset

- a. The developer as a Team Lead (The Mindset Shift)
- b. What is spec-first development
- c. Prompt engineering and context managing

3. Building project using ai-first approach (part 2): Core Development

- a. Set up development environment
- b. Setting up agent rules
- c. Development loop with agent

4. Testing the project using agent testing (part 3): CI & Feedback Loop

- a. Project test coverage
- b. Project ci/cd configuration (via GitHub Actions)
- c. Agent feedback loop (reading CI logs)

5. Project QA and security risks (part 4): Review & Retrospective

- a. Code responsibility and human code review
- b. Security risks with agents
- c. Human Retrospective: Improving the Spec & Prompting Process

6. Deep dive into the agent eco-system

- a. Popular models and their strength
- b. Compare between the popular agent tools
- c. Using advanced features: Sub-agents, MCP, and Claude Code plugins

7. Developer as an agents team lead

- a. Specification, Criticism, Integration, management
- b. AI friendly architecture
- c. Orchestrating team of agents

8. Advance practice with real world project

- a. Model context limitation (limitation on large projects)
- b. Refactoring existing code
- c. Adding test coverage to existing code
- d. Adding features to existing code

9. Keep calm and catch up

- a. How to stay updated
- b. How the future will look
- c. Q&A